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A thermodynamic assessment of the Mg–Sn–Sb ternary system

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ABSTRACT

Magnesium-tin-antimony (Mg–Sn–Sb) system is a promising system for developing high-creep resistance magnesium alloy and the alloy-type anode of magnesium recharge battery. Herein, the thermodynamic assessment of Mg–Sn–Sb ternary system was carried out using the CALculaiton of PHAse Diagram (CALPHAD) approach based on available literature data. The liquid phase was described *via* the associated model with associates Mg_3Sb_2 and Mg_2Sn, and the compounds were described using the sublattice model. The consistency of the calculated phase equilibria relations and thermochemical properties with the available experimental data of Mg–Sn–Sb ternary system is presented. The thermodynamic descriptions of Mg–Sb, Sn–Sb binary systems and Mg–Sn–Sb ternary system assessed in this work provide a thermodynamic basis for the alloy design of magnesium alloy.

1. Introduction

Magnesium (Mg) alloys as the light-weight structure materials have attracted many interests in the automotive and aeronautic industries owing to their good damping capacity and environmentally friendly characteristics [1–4]. However, the applications of Mg alloys have not been widely adopted due to their inferior corrosion resistance, poor creep-resistance and unsatisfactory formability [5–8]. Many studies have been performed to design advanced-performance Mg alloys by the addition of alloying elements and optimization of subsequent processes.

The high-temperature properties of Mg alloys are possible to be improved *via* developing Mg alloys containing the high-temperature-stable intermetallics. Bi, Sn, Si, Sb and rare earth (RE) elements are the promising alloying elements [9–24]. Specifically, magnesium-tin (Mg–Sn) alloys have, recently, attracted a lot of attentions [13–18]. And, magnesium-antimony (Mg–Sb) system is also an interesting system to develop high-temperature Mg alloys [15–19], because the addition of Sb causes the precipitation of rod-shaped α-Mg_3Sb_2, which can strengthen the Mg-matrix and the grain boundaries. In addition, the addition of Sb can make the precipitated Mg_2Sn phase being more homogeneous distribution within the Hcp(Mg) [15–17,24], which is possible to enhance the mechanical properties of the Mg–Sn alloys. Besides, the magnesium-tin-antimony (Mg–Sn–Sb) ternary system is also a promising candidate material for development of novel alloy-type anodes for rechargeable magnesium batteries (RMBs) [25–27].

Therefore, it is crucial to carry out a critical-evaluation and optimization of the phase diagram and thermodynamic property of the Mg–Sn–Sb ternary system [28,29]. In this work, self-consistent and credible thermodynamic parameters of the Mg–Sn–Sb system were acquired, where the Mg–Sb and Sn–Sb binary systems were re-optimized by present authors and the thermodynamic description of Mg–Sn system was adopted from Meng et al. [30].

2. Literature review

The following sections critically assessed the studies on Mg–Sn–Sb ternary system and Mg–Sn, Mg–Sb, Sn–Sb binary sub-systems in the previous publications.

2.1. The Mg–Sn binary system

The Mg–Sn binary system has been critically reviewed and thermodynamically assessed by several groups [30–33]. The Mg_2Sn phase, with a narrow homogeneous range, is the only stable intermetallic phase in Mg–Sn system with a congruent melting point at about 770 °C. In addition, strong short range ordering was observed in the liquid phase, thus Mg_2Sn associates was proposed for the liquid phase modeling [31]. As the solubility of Sn in the Hcp(Mg) phase was not considered in Ref. [32], and only part of the parameters were provided in Ref. [33], which could not repeat the calculations. And we accepted the thermodynamic description of Mg–Sn system, where the strong short range ordering behavior of the liquid phase in this binary system can be successfully described using the associate solution model, provided by Meng et al. [30] as a start point of this work. The calculated Mg–Sn

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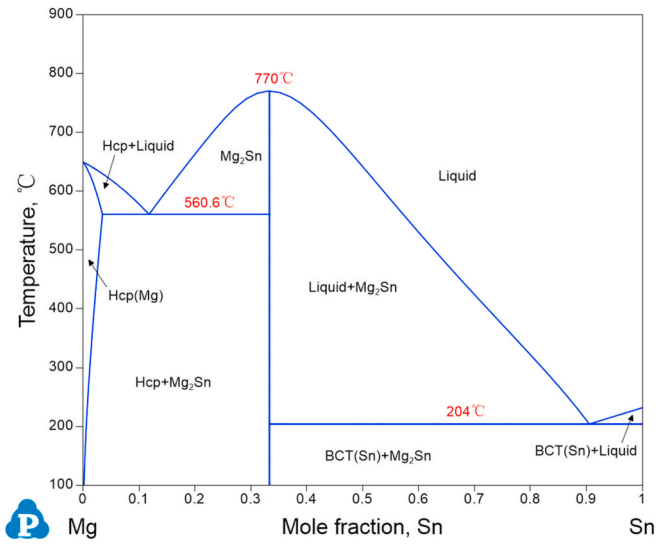


Fig. 1. Calculated phase diagram of Mg–Sn system [30].

binary phase diagram obtained from Ref. [30] is shown in Fig. 1, and their calculated results are in good agreement with most of the experimental data, which will be discussed later.

2.2. The Mg–Sb binary system

Several thermodynamic assessments of the Mg–Sb system have been published [34–38]. Jonsson and Agren [35] used two-sublattice ionic model for the liquid phase and treated the α-Mg_3Sb_2 phase as a stoichiometric compound. In addition, as it was published before 1991, the lattice stability parameters for the pure elements are not from the later generally accepted Dinsdale data [39]. Balakumar and Medraj [36] modeled the liquid phase as substitutional solution model with Redlich-Kister polynomial formula but produced inverted miscibility gap at high temperature. Besides, the congruent melting point of β-Mg_3Sb_2 phase could not be accurately reproduced at about 1228 °C reported in Ref. [34] and the α-Mg_3Sb_2 phase was also considered as a stoichiometric compound in their re-optimization [36]. Later, Malakhov and Balakumar [37] tried to eliminate the inverted miscibility gap by introducing complicated topological constraints, which are difficult to apply to multi-component systems. Afterwards, Paliwal and Jung [38] re-assessed the system using the modified quasi-chemical model (MQM), which can only be used in FactSage software, to describe the short range ordering behavior of liquid phase. Therefore, we re-assessed the Mg–Sb binary system based on the available literature data reported in Refs. [34,40–52] in the present work, using the associate solution model with Mg_3Sb_2 associate for the liquid, as similar to the Mg–Sn binary system.

2.3. The Sn–Sb binary system

The experimental work on Sn–Sb system before 2017 has been systematically reviewed by Schmetterer et al. [53]. By then, two versions of Sn–Sb phase diagrams were widely used. One version in Ref. [54] was based on the calculation from Ref. [55], and the other one was reported in the ASM Handbook on Binary Alloy Phase Diagrams as reviewed in Ref. [56] based on the experimental data in Ref. [57]. In both versions [54,56], the Sn_3Sb_2 phase was stable phase but with different stable temperature range. More recently, Schmetterer et al. [58] reported a new Sn–Sb phase diagram based on the new experimental results from XRD, SEM/EDX and DTA. Their reported results [58] revealed that the new Sn_4Sb_3 phase with $R\bar{3}m$ space group is the stable phase rather than the Sn_3Sb_2 phase without clear structural description, which was accepted in the widely used versions [54,56] of this binary phase diagram before then. The stability of the Sn_4Sb_3 phase was further confirmed by the ab initio calculation [59], where the calculated unit cell size for relaxed structure was in general accord with the size of a unit cell of Sn_4Sb_3 phase from XRD results reported by Schmetterer et al. [58]. The special attention focused on the phase equilibria within the composition range of Sn_4Sb_3–SnSb was examined by DSC and metallograph combined with SEM/EDS [60], and the phase transformation information related to the Sn_4Sb_3 phase was also reconfirmed.

The Sn–Sb system was modeled by Ohtani et al. [61], Jonsson and Agren [62] and Chen et al. [55], respectively. Recently, Gierlotka [59] remodeled the Sn–Sb system based on the new version of this binary experimental phase diagram data established from the XRD, EDX and DTA results [58] and their own ab initio calculation for the enthalpy formation of Sn_4Sb_3 phase. In ref. [59], the exponential function was used for the interaction parameters of the liquid phase. Except to avoid the inverted miscibility gap at high temperature, the exponential function could not provide any more benefits than the traditional linear equation, but increase the risk of low-T stable artifacts [63,64]. In this system, the linear equation can be used to describe the liquid phase and without showing the inverted miscibility gap as another assessment reported by Lysenko [65]. The disadvantage of the thermodynamic description in Ref. [65] is that they accepted the contradicted version of phase diagrams including both the Sn_4Sb_3 and Sn_3Sb_2 phases, which is against the experimental results in Refs. [58,60], as their dedicated experimental results agree with each other and confirmed that the Sn_4Sb_3 phase rather than the Sn_3Sb_2 phase is the stable phase at the Sn rich side of the Sn–Sb system [58,60].

Therefore, the Sn–Sb binary system was re-modeled in this work based on the available literature data in Refs. [53–61,66–76] using substitutional solution model with linear equations for better compatibility with other thermodynamic description in the Mg-based alloys. The experimental information reported by Predel et al. [57], Ohtani et al. [61], Chen et al. [55], Schmetterer et al. [58] and Borzone et al. [60] were given high weights in the optimization work.

2.4. The Mg–Sn–Sb ternary system

The Mg–Sn–Sb ternary system has been reviewed by Kinzhybalo [77] based on the experimental data [51,78–80] and the thermodynamic calculation [81]. Bolshakov et al. [78] experimentally measured the pseudobinary section of Mg_3Sb_2–Mg_2Sn by thermal analysis, X-ray diffraction and optical microstructure. They [78] determined the variation in the lattice parameter of Mg_2Sn phase when adding Sb by X-ray analysis and proposed the β-Mg_3Sb_2 phase and Mg_2Sn phase formed a solid solution over the whole concentration range at high temperature based on their limited experimental results. But because of the different structures between the β-Mg_3Sb_2 phase (structure type $D5_3$) and Mg_2Sn phase (structure type CaF_2), Jonsson and Agren treated them as two different phases in the thermodynamic modeling [81], Kinzhybalo also believed in the review [77] that a complete solid solution is not possible between these two phases. On the other hand, Jung et al. [14] modeled these two phases as one phase with solid solution range based on the experimental results from Ref. [78], however, they did not provide their calculated results on this pseudobinary phase diagram.

In the present work, we accept Kinzhybalo's review [77] and the modeling used Jonsson and Agren [81], treat the β-Mg_3Sb_2 and Mg_2Sn phase as two different phases. This structure difference of these two phases will be discussed below in detail. Nevertheless, the thermodynamic description of Mg–Sn–Sb ternary system has been modeled in 1987 [81], many updates are required for the binary subsystems, and even the lattice stability for the pure elements are also updated [39]. Therefore, Mg–Sn–Sb system is reassessed based on the updated binary subsystems. The experimental information on the phase relations in Refs. [78,80] as well as thermodynamic data in Ref. [51] were taken into account in the present optimization.

Table 1
The crystal structures of the phases in the Mg–Sn–Sb ternary system.

phase	Pearson Symbol	Prototype	Space group	Lattice Parameters, nm		Reference
				A	c	
α(Mg)	*hP2*	Mg	$P6_3/mmc$	0.32093	0.52107	[41]
α(Sn)	*cF8*	C (diamond)	$Fd\bar{3}m$	0.64892	…	[31]
β(Sn)	*tI4*	βSn	$I4_1/amd$	0.58316	0.31815	[41]
α(Sb)	*hR2*	αAs	$R\bar{3}m$	0.45065	α = 57.11°	[41]
β-Mg_3Sb_2	*cI80*	Mn_2O_3	$Ia\bar{3}$	…	…	[34]
α-Mg_3Sb_2	*hP5*	La_2O_3	$P\bar{3}m1$	0.45822	0.72436	[34]
Mg_2Sn	*cF12*	CaF2	$Fm\bar{3}m$	0.675944	…	[84]
SnSb	*rhombohedral*	NaCl-derivative	$R\bar{3}m$	0.3114	α = 86.4°	[58]
Sn_4Sb_3	*rhombohedral*	Sn_4As_3	$R\bar{3}m$	0.433111	3.7302	[58]

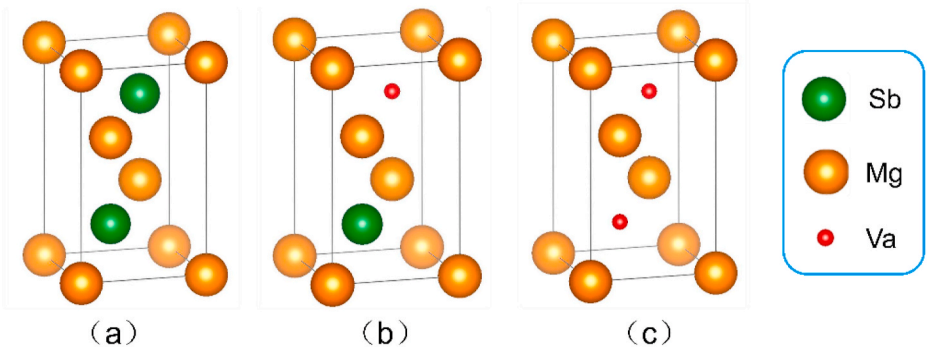


Fig. 2. Illustration of α-Mg_3Sb_2-D52 structure: (a) no Sb site occupied by Va; (b) partial Sb sites occupied by Va; (c) all Sb sites occupied by Va.

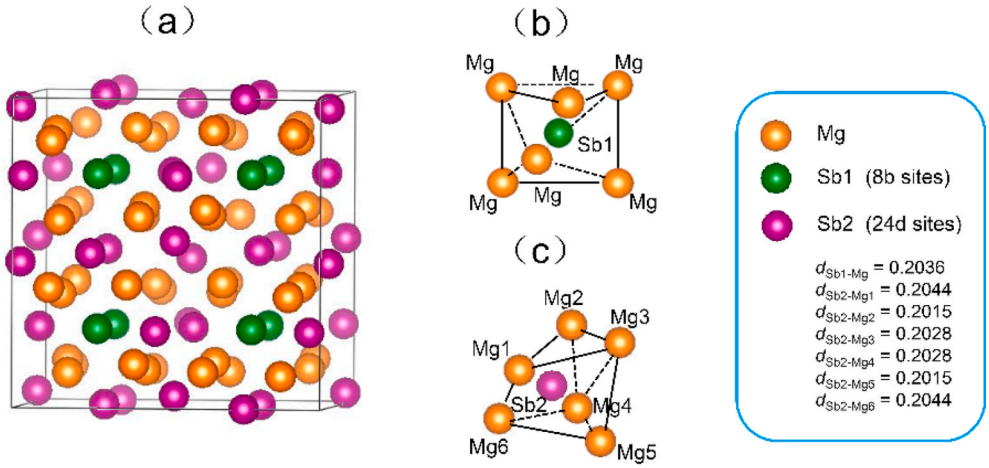


Fig. 3. Illustration of β-Mg_3Sb_2-$D5_3$ structure: (a) structure of β-Mg_3Sb_2; (b) the nearest neighbor coordinates of Sb1 (8b sites); (c) the nearest neighbor coordinates of Sb2 (24d sites).

3. Thermodynamic model

3.1. Pure elements

The Gibbs energy for the element i (i = Mg, Sn and Sb) in the phase Φ, Φ as liquid, hexagonal close-packed (hcp(Mg)), Bct(Sn) and Rhombohedral(Sb), is described by an equation of the following form:

$$ {}^{0}G_i^{\varphi}(T) = G_i^{\varphi}(T) - H_i^{SER}(298.15K) = a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^{-9} \quad (1) $$

where $H_i^{SER}(298.15K)$ is the enthalpy of the element i in its standard reference state (SER); T is the absolute temperature in K; $G_i^{\varphi}(T)$ is the Gibbs energy of the element i with structure Φ; ${}^{0}G_i^{\varphi}(T)$ is the molar Gibbs energy of the element i with the structure of Φ referred to the enthalpy of SER, and the thermodynamic parameters a to h are taken from SGTE (Scientific Group Thermodata Europe). In the present work, ${}^{0}G_{Mg}^{\varphi}(T)$, ${}^{0}G_{Sn}^{\varphi}(T)$and ${}^{0}G_{Sb}^{\varphi}(T)$ are adopted from the SGTE database, https://www.sgte.net/en/free-pure-substance-database, in which the information of element Sn has been updated [39].

3.2. Liquid phase

In the present work, the liquid phase is described using associate solution model [82]. Two associates, Mg_2Sn and Mg_3Sb_2, are introduced

Table 2
Thermodynamic parameters of the Mg–Sn–Sb ternary system.

Phase, model	Thermodynamic parameters[a]	Reference
Liquid, (Mg, Mg_3Sb_2, Mg_2Sn, Sb, Sn)	${}^0G^L_{Mg_2Sn} = -66092.9 + 94.809\cdot T - 11.576\cdot T\cdot LnT + 2\cdot {}^0G^L_{Mg} + {}^0G^L_{Sn}$	[30]
	${}^0L^L_{Mg_2Sn,Sn} = -12468.2 - 4.815\cdot T$	[30]
	${}^0L^L_{Mg,Sn} = -30841.1 + 0.781\cdot T$	[30]
	${}^0L^L_{Mg,Mg_2Sn} = +5970.6 - 9.444\cdot T$	This work
	${}^0G^L_{Mg_3Sb_2} = -261367 + 125.8\cdot T - 9.85\cdot T\cdot LnT + 3\cdot {}^0G^L_{Mg} + 2\cdot {}^0G^L_{Sb}$	This work
	${}^0L^L_{Mg,Mg_3Sb_2} = -14941.2 - 14.3663\cdot T$	This work
	${}^1L^L_{Mg,Mg_3Sb_2} = +10399.95 + 3.0\cdot T$	This work
	${}^0L^L_{Mg_3Sb_2,Sb} = -19774.6 - 21.997\cdot T$	This work
	${}^1L^L_{Mg_3Sb_2,Sb} = -17802.3 + 3.3426\cdot T$	This work
	${}^0L^L_{Mg,Sb} = -64574.5 - 13.8\cdot T$	This work
	${}^1L^L_{Mg,Sb} = -1.0\cdot T$	This work
	${}^0L^L_{Sb,Sn} = -5800 - 2.15\cdot T$	This work
	${}^1L^L_{Sb,Sn} = -200 - 1.45\cdot T$	This work
	${}^2L^L_{Sb,Sn} = 2350$	This work
	${}^0L^L_{Mg_3Sb_2,Mg_2Sn} = -4561.62 + 9.0907\cdot T$	This work
	${}^1L^L_{Mg_3Sb_2,Mg_2Sn} = +4.0\cdot T$	This work
	${}^0L^L_{Mg_3Sb_2,Sn} = -37486 + 25.6327\cdot T$	This work
	${}^0L^L_{Mg_2Sn,Sb} = -44990 - 25\cdot T$	This work
	${}^1L^L_{Mg_2Sn,Sb} = -33993$	This work
	${}^0L^L_{Mg,Mg_3Sb_2,Sn} = -24998.0 - 15.0\cdot T$	This work
	${}^0L^L_{Mg,Mg_2Sn,Sb} = -25782.0$	This work
	${}^0L^L_{Mg,Sb,Sn} = -26649.7 - 25.3526\cdot T$	This work
	${}^1L^L_{Mg,Sb,Sn} = -20335.3 - 25.6582\cdot T$	This work
Bct(Sn), (Sb, Sn)	${}^0L^{Bct}_{Sb,Sn} = -5200 + 12.5\cdot T$	This work
Rhombohedral(Sb), (Sb, Sn)	${}^0L^{Rhombohedral}_{Sb,Sn} = +6400 - 5\cdot T$	This work
	${}^1L^{Rhombohedral}_{Sb,Sn} = -5000$	This work
Hcp(Mg), (Mg, Sb, Sn)	${}^0L^{Hcp}_{Mg,Sb} = -61998.6 - 18.999\cdot T$	This work
	${}^0L^{Hcp}_{Mg,Sn} = -26256.5 + 7.234\cdot T$	[30]
	${}^1L^{Hcp}_{Mg,Sn} = -25895.7$	This work
	${}^2L^{Hcp}_{Mg,Sn} = -7030$	This work
	${}^0L^{Hcp}_{Mg,Sb,Sn} = 59997 + 39\cdot T$	This work
α-Mg_3Sb_2, Mg_3(Sb, Sn, Va)$_2$	${}^0G^{\alpha-Mg_3Sb_2}_{Mg:Sb} = -294509 - 14.05\cdot T + 11.212\cdot T\cdot LnT - 0.013\cdot T^2 + 3\cdot {}^0G^{Hcp}_{Mg} + 2\cdot {}^0G^{Rhombohedral}_{Sb}$	This work
	${}^0G^{\alpha-Mg_3Sb_2}_{Mg:Va} = +40665.1 + 9\cdot T + 3\cdot {}^0G^{Hcp}_{Mg}$	This work
	${}^0L^{\alpha-Mg_3Sb_2}_{Mg:Sb,Va} = -13675.1 + 5.432\cdot T$	This work
	${}^1L^{\alpha-Mg_3Sb_2}_{Mg:Sb,Va} = -14000 + 7.476\cdot T$	This work
	${}^0G^{\alpha-Mg_3Sb_2}_{Mg:Sn} = -62584.5 - 11.8001\cdot T + 3\cdot {}^0G^{Hcp}_{Mg} + 2\cdot {}^0G^{Bct}_{Sn}$	This work
	${}^0L^{\alpha-Mg_3Sb_2}_{Mg:Sb,Sn,Va} = -16000 + 7.9855\cdot T$	This work
β-Mg_3Sb_2, Mg_6(Sb, Sn, Va)$_3$(Sb, Sn)$_1$	${}^0G^{\beta-Mg_3Sb_2}_{Mg:Sb:Sb} = -386039 - 190.64\cdot T + 17.16\cdot T\cdot LnT + 6\cdot {}^0G^{Hcp}_{Mg} + 4\cdot {}^0G^{Rhombohedral}_{Sb}$	This work
	${}^0G^{\beta-Mg_3Sb_2}_{Mg:Va:Sb} = -33000.4 - 15.1\cdot T + 14.25\cdot T\cdot LnT + 6\cdot {}^0G^{Hcp}_{Mg} + 1\cdot {}^0G^{Rhombohedral}_{Sb}$	This work
	${}^0L^{\beta-Mg_3Sb_2}_{Mg:Sb,Va:Sb} = -68212.8 - 29.878\cdot T$	This work
	${}^1L^{\beta-Mg_3Sb_2}_{Mg:Sb,Va:Sb} = -106813 - 28.0\cdot T$	This work
	${}^0G^{\beta-Mg_3Sb_2}_{Mg:Sb:Sn} = -84242 - 84.7975\cdot T + 6\cdot {}^0G^{Hcp}_{Mg} + 3\cdot {}^0G^{Rhombohedral}_{Sb} + 1\cdot {}^0G^{Bct}_{Sn}$	This work
	${}^0G^{\beta-Mg_3Sb_2}_{Mg:Sn:Sb} = -60000 - 47.7051\cdot T + 6\cdot {}^0G^{Hcp}_{Mg} + 3\cdot {}^0G^{Bct}_{Sn} + 1\cdot {}^0G^{Rhombohedral}_{Sb}$	This work
	${}^0G^{\beta-Mg_3Sb_2}_{Mg:Sn:Sn} = -68551.5 - 129.965\cdot T + 6\cdot {}^0G^{Hcp}_{Mg} + 4\cdot {}^0G^{Bct}_{Sn}$	This work
	${}^0G^{\beta-Mg_3Sb_2}_{Mg:Va:Sn} = -33966.8 + 31.2632\cdot T + 6\cdot {}^0G^{Hcp}_{Mg} + 1\cdot {}^0G^{Bct}_{Sn}$	This work

(*continued on next page*)

Table 2 (*continued*)

Phase, model	Thermodynamic parameters[a]	Reference
	${}^{0}L^{\beta-Mg_3Sb_2}_{Mg:Sb:Sb,Sn} = {}^{0}L^{\beta-Mg_3Sb_2}_{Mg:Sn:Sb,Sn} = {}^{0}L^{\beta-Mg_3Sb_2}_{Mg:Va:Sb,Sn} = -115022 + 10.0{\cdot}T$	
	${}^{0}L^{\beta-Mg_3Sb_2}_{Mg:Sb,Sn:Sb} = +99194.1 - 19.3952{\cdot}T$	This work
	${}^{0}L^{\beta-Mg_3Sb_2}_{Mg:Sb,Sn:Sn} = -239404 - 31.5356{\cdot}T$	This work
	${}^{0}L^{\beta-Mg_3Sb_2}_{Mg:Va,Sn:Sn} = {}^{0}L^{\beta-Mg_3Sb_2}_{Mg:Va,Sn:Sb} = -50170.2 + 11.7556{\cdot}T$	This work
	${}^{0}L^{\beta-Mg_3Sb_2}_{Mg:Sb,Va:Sn} = -47732.8 - 25.1993{\cdot}T$	This work
Mg_2Sn, (Mg, Va)$_2$(Sb, Sn)	${}^{0}G^{Mg_2Sn}_{Mg:Sn} = -96165.9 + 339.999{\cdot}T - 66.285{\cdot}T{\cdot}LnT + 96000{\cdot}T^{-1} - 0.012166{\cdot}T^2 + 3.33828{\cdot}10^{-7}{\cdot}T^3$	[30]
	${}^{0}G^{Mg_2Sn}_{Mg:Sb} = -31803.7 - 48.8857{\cdot}T + 2{\cdot}{}^{0}G^{Hcp}_{Mg} + {}^{0}G^{Rhombohedral}_{Sb}$	This work
	${}^{0}G^{Mg_2Sn}_{Va:Sb} = +18216.8 + 6.9678{\cdot}T + {}^{0}G^{Rhombohedral}_{Sb}$	This work
	${}^{0}G^{Mg_2Sn}_{Va:Sn} = +14395.7 + 3.7217{\cdot}T + {}^{0}G^{Bct}_{Sn}$	This work
	${}^{0}L^{Mg_2Sn}_{Mg:Sb,Sn} = -43163.3 - 15.2118{\cdot}T$	This work
	${}^{1}L^{Mg_2Sn}_{Mg:Sb,Sn} = -5000 - 20{\cdot}T$	This work
	${}^{0}L^{Mg_2Sn}_{Va:Sb,Sn} = -7000 + 4.7809{\cdot}T$	This work
	${}^{0}L^{Mg_2Sn}_{Mg,Va:Sb} = -74223.6 - 21.9598{\cdot}T$	This work
	${}^{0}L^{Mg_2Sn}_{Mg,Va:Sn} = +10077$	This work
	${}^{1}L^{Mg_2Sn}_{Mg,Va:Sn} = +8000$	This work
SnSb, (Sn, Sb, Va)$_4$Sb$_3$	${}^{0}G^{SnSb}_{Sn:Sb} = +390 + 8.5{\cdot}T + {}^{0}G^{Sn_4Sb_3}_{Sn:Sb}$	This work
	${}^{0}G^{SnSb}_{Sb:Sb} = +88000 + 42{\cdot}T + 7{\cdot}{}^{0}G^{Rhombohedral}_{Sb}$	This work
	${}^{0}G^{SnSb}_{Va:Sb} = 12100 - 2{\cdot}T + 3{\cdot}{}^{0}G^{Rhombohedral}_{Sb}$	This work
	${}^{0}G^{SnSb}_{Sn,Sb,Va:Sb} = +86000 + 25.5{\cdot}T$	This work
	${}^{1}G^{SnSb}_{Sn,Sb,Va:Sb} = -48000$	This work
Sn_4Sb_3, Sn_4Sb_3	${}^{0}G^{Sn_4Sb_3}_{Sn:Sb} = -29640 + 5.2{\cdot}T + 4{\cdot}{}^{0}G^{Bct}_{Sn} + 3{\cdot}{}^{0}G^{Rhombohedral}_{Sb}$	This work

[a] Gibbs energies are expressed in J/mol. All lattice stabilities of Mg, Sn and Sb are given by SGTE.

Table 3
Invariant reactions of the Mg–Sn system.

Reaction	T (°C)	Composition, x_{Sn}			Method	Reference
L ↔ Hcp	564.8	0.116	…	0.3333	Experiment	[87]
(Mg) +	567	0.107	0.0350	…	Experiment	[91]
Mg_2Sn	561.2	0.11	…	…	Experiment	[88]
	561	0.105	…	…	Experiment	[89]
	560.6	0.107	0.0335	…	Experiment	[92]
	560.2	0.118	0.035	0.3333	Calculated	[30]
	561.1	0.122	0.0398	0.3333	Calculated	This work
L ↔ Mg_2Sn	783	0.3333	0.3333	…	Experiment	[87]
	774	…	…	…	Experiment	[90]
	770.5	…	…	…	Experiment	[88]
	778	…	…	…	Experiment	[89]
	770	…	…	…	Experiment	[50]
	770	0.3333	0.3333	…	Calculated	[30]
	771	0.3333	0.3333	…	Calculated	This work
L ↔ Mg_2Sn	197.5	0.914	0.3333	1	Experiment	[91]
+ Bct(Sn)	203.5	0.904	…	…	Experiment	[88]
	209.4	0.889	…	…	Experiment	[87]
	204.3	0.905	0.3333	1	Calculated	[30]
	204.4	0.906	0.3367	1	Calculated	This work

based on the experimental results about the short range ordering in the liquid phase of Mg–Sn and Mg–Sb systems. The molar Gibbs energy of the liquid phase is described by:

$$G^L_m(T) - H^{SER}(298.15K) = \sum_i x_i {}^{0}G^L_i(T) + RT\sum_i x_i \ln x_i + {}^{E}G^L_m \quad (2)$$

where x_i and ${}^{0}G^L_i(T)$ (i = Mg, Sn, Sb, Mg_2Sn and Mg_3Sb_2) are the mole fraction and the molar Gibbs energy of the pure species i in the liquid state and ${}^{E}G^L_m$ is the excess Gibbs energy. ${}^{0}G^L_{Mg_2Sn}$, ${}^{0}G^L_{Mg_3Sb_2}$ and ${}^{E}G^L_m$ can be described using the associate solution model by the following

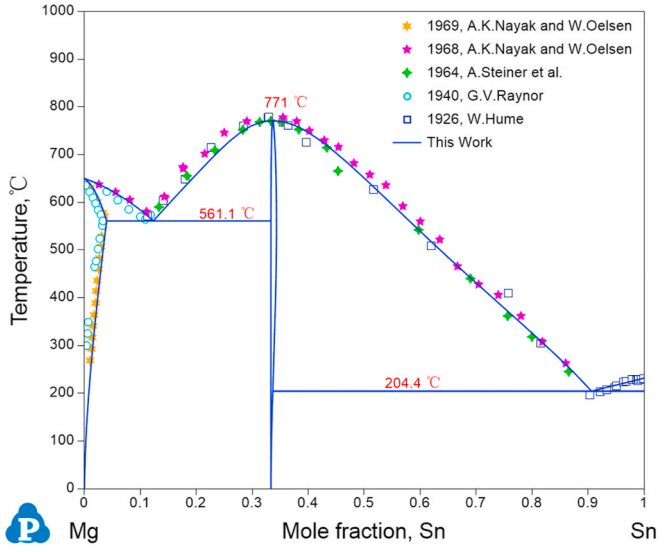


Fig. 4. Calculated phase diagram of the Mg–Sn system in the present work, compared with the experimental data compiled in Ref. [31].

expressions, respectively:

$${}^{0}G^L_{Mg_2Sn} = a_1 + b_1T + c_1T\ln T + 2{}^{0}G^L_{Mg} + {}^{0}G^L_{Sn} \quad (3)$$

$${}^{0}G^L_{Mg_3Sb_2} = a_2 + b_2T + c_2T\ln T + 3{}^{0}G^L_{Mg} + 2{}^{0}G^L_{Sb} \quad (4)$$

Table 4
Invariant reactions of the Mg–Sb system.

Reaction	T (°C)	Composition, x_{Sb}			Method	Reference
L ↔ Mg	649	0	0	…	Experiment	[41]
L ↔ Hcp(Mg)	627	0.11	~0.0	0.4	Experiment	[42]
+ α-Mg_3Sb_2	629	0.1	~0.0	0.385	Experiment	[43]
	630	~0.0	~0.0	0.4	Experiment	[44]
	627	0.03	0	0.38	Calculated	[38]
	628.54	0.047	0.001	0.379	Calculated	This work
β-Mg_3Sb_2 ↔ L	894	0.392	0.335	0.397	Experiment	[43]
+ α-Mg_3Sb_2	891	0.38	0.295	0.389	Calculated	[38]
	893.6	0.384	0.315	0.393	Calculated	This work
L ↔ β-Mg_3Sb_2	1228	0.4	0.4	…	Experiment	[43]
	1227	0.4	0.4	…	Calculated	[38]
	1228	0.399	0.399	…	Calculated	This work
β-Mg_3Sb_2 ↔	930	0.4	0.4	…	Experiment	[43]
α-Mg_3Sb_2	925	0.4	0.4	…	Experiment	[34]
	929	0.4	0.4	…	Calculated	[38]
	930.7	0.4	0.4	…	Calculated	This work
L + β-Mg_3Sb_2	930	0.456	0.4	0.401	Calculated	[34]
↔ α-Mg_3Sb_2	929	0.489	0.4	0.4	Calculated	[38]
	930.7	0.471	0.399	0.4	Calculated	This work
L ↔ Sb	630.75	1	1	…	Experiment	[41]
L ↔ α-Mg_3Sb_2	594	0.794	0.4	1	Experiment	[42]
+ Rhom(Sb)	591	0.791	0.4	1	Experiment	[40]
	579	0.86	0.4	1	Experiment	[43]
	580	0.863	0.4	1	Calculated	[38]
	579.9	0.858	0.4	1	Calculated	This work

$$ {}^{E}G_m^L = \sum_i \sum_{j>i} x_i x_j \sum_{v=0}^{n} {}^{v}L_{i,j}^{L}(x_i - x_j)^v + \sum_i \sum_{j>i} \sum_{k>j} x_i x_j x_k ({x_i}^0 L_{i,j,k}^{L} + {x_j}^1 L_{i,j,k}^{L} + {x_k}^2 L_{i,j,k}^{L}) \quad (5) $$

where, ${}^{v}L_{i,j}^{L}$ ($v = 0, 1, 2$ and i, j = Mg, Sn, Sb, Mg_2Sn and Mg_3Sb_2) is the interaction parameters of the species i and j, and ${}^{v}L_{i,j,k}^{L}$ ($v = 0, 1, 2$ and i, j, k = Mg, Sn, Sb, Mg_2Sn and Mg_3Sb_2) is the interaction parameters of the species i, j and k.

3.3. Solid solution phases

The substitution solution model is chosen to describe the solid solution phases Φ: Hcp(Mg), Bct(Sn) and Rhombohedral(Sb). The molar Gibbs energies of Φ phases are described as:

$$ G_m^{\varphi} = \sum {x_i}^0 G_i^{\varphi} + RT \sum x_i \ln x_i + x_i x_j \sum_{v=0}^{n} {}^{v}L_{i,j}^{\varphi}(x_i - x_j)^v \quad (6) $$

where R is the gas constant, x_i and x_j (i, j = Mg, Sn, Sb) are the mole fractions. ${}^{v}L_{Mg,Sn}^{Hcp}$, ${}^{v}L_{Mg,Sb}^{Hcp}$, ${}^{v}L_{Sb,Sn}^{Bct}$ and ${}^{v}L_{Sb,Sn}^{Rhombohedral}$ ($v = 0, 1, 2$) are binary interaction parameters in the Mg–Sn, Mg–Sb and Sn–Sb binary systems, respectively. The ternary interaction parameters are not given for the solid solution Φ phases, except Hcp(Mg), because of no experimental information in this part.

3.4. Intermetallic compounds

The Compound Energy Formalism (CEF) proposed by Mats Hillert [83] is used to describe the Gibbs energy of the intermetallic phases. As no ternary intermetallic phase is reported in Mg–Sn–Sb ternary system, five binary intermetallic compounds, Mg_2Sn in the Mg–Sn system, α-Mg_3Sb_2 and β-Mg_3Sb_2 in the Mg–Sb binary system, SnSb and Sn_4Sb_3 in the Sn–Sb binary system, are modeled in this work based on their crystal structures. The crystal structures of these binary phases are listed in Table 1. The Gibbs energy is described as per mole of formula unit with the expression as below:

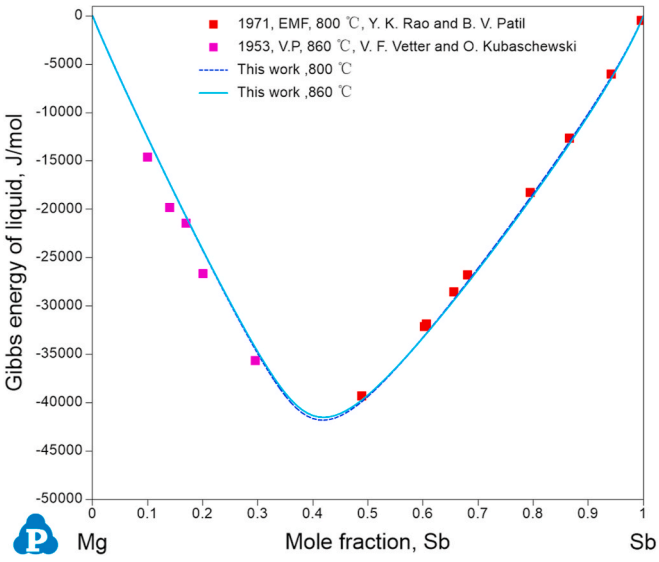


Fig. 6. Calculated molar Gibbs free energies of the liquid phase of the Mg–Sb system at 800 °C and 860 °C compared with measured data [45,49], (Reference state: liquid Mg and liquid Sb).

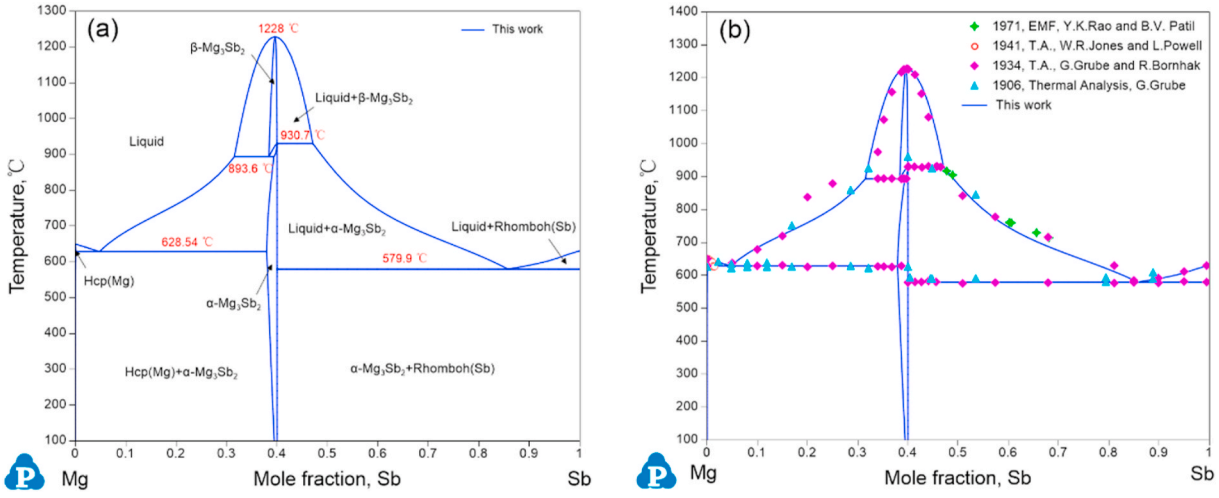


Fig. 5. Calculated phase diagram of the Mg–Sb system in the present work, compared with the experimental data [42–45].

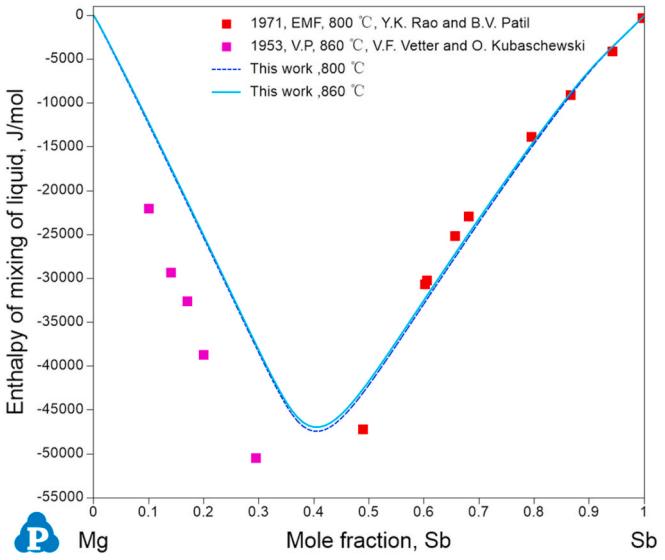


Fig. 7. Calculated molar enthalpy of mixing of the liquid phase of the Mg–Sb at 800 and 860 °C compared with measured data [45,49], (Reference state: liquid Mg and liquid Sb).

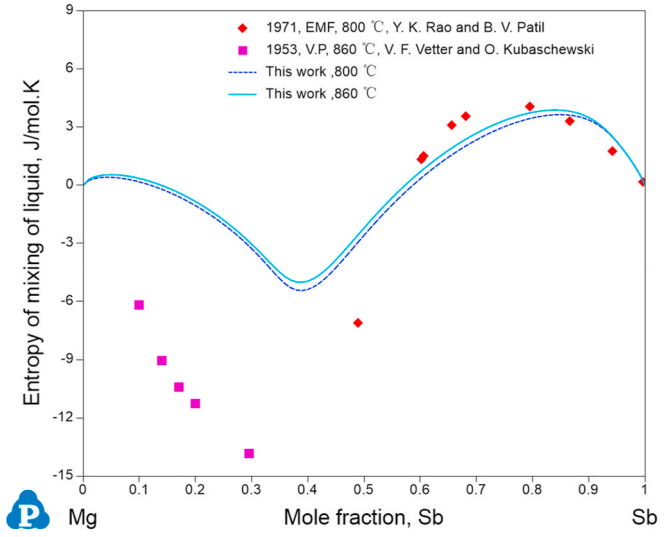


Fig. 8. Calculated molar entropy of mixing of the liquid phase of the Mg–Sb at 800 and 860 °C compared with measured data [45,49], (Reference state: liquid Mg and liquid Sb).

$$G_m - \sum n^s(1-y^s_{Va})\sum x^0_I G^{stst}_I = \sum\sum \Delta^0_f G_{end}\prod y^s_J + RT\sum\sum n^s y^s_J \ln(y^s_J) + {}^{E}G_m \quad (7)$$

where, the left hand side represents the mixing Gibbs energy, ${}^{M}G_m$. And numerical values of ${}^{0}G_{end}$ were given relative to standard states (*stst*) of the components *I*. ${}^{E}G_m$is regarded as excess terms.

3.4.1. Mg_2Sn phase

In the Mg–Sn binary system, the Mg_2Sn phase was considered as a stoichiometric compound because of its narrow homogeneity range [30]. However, Bolshakov et al. [78] reported that Mg_2Sn phase and the β-Mg_3Sb_2 phase could form a solid solution over the whole concentration range in the Mg_2Sn–Mg_3Sb_2 vertical section. In the present work, we believe the existence of some solid solubility of Sb in Mg_2Sn phase to be possible rather than the complete solid solution phase. The crystallographic structure of Mg_2Sn belongs to the CaF_2 prototype, which has two sublattice occupations: Mg (8c) at (0.25, 0.25, 0.25) and Sn (4a) at (0, 0, 0) [84]. And, the Sb atoms may replace the either sublattice or both sites to form a certain solubility. Specifically, Bolshakov et al. [78] reported that on complete replacement of Sn by Sb, it is necessary to introduce 2 mol of neutral vacancies (Va) for per 1 mol of Sb in crystal. Therefore, the ternary Mg_2Sn phase was regarded as $(Mg, Va)_2(Sb, Sn)$ in order to produce reasonable solid solubility of Sb in Mg_2Sn phase. In the present work, this lattice description of the Mg_2Sn phase bring minor changes in the parameters of the Mg–Sn binary system provided by Meng et al. [30]. Thus, the sub-system parameters have also been fine-tuned to ensure the fitting of phase diagram and thermodynamic properties of this system.

3.4.2. α-Mg_3Sb_2 phase

As shown in Fig. 2, the intermetallic compound α-Mg_3Sb_2, which crystallizes in the La_2O_3 prototype [85], has a homogeneity range. In the crystal structure of the phase α-Mg_3Sb_2, three kinds of atoms are located in the unit cell: Sb (2d) at (0.333, 0.667, 0.222), Mg1 (1a) at (0, 0, 0) and Mg2 (2d) at (0.333, 0.667, 0.628). Considering a small amount of excess Mg of the non-stoichiometric compound and the structural feature of the atom site occupation, the α-Mg_3Sb_2 phase was treated as the formula $Mg_3(Sb, Va)_2$ with two sublattices. Fig. 2(b) and c shows the situations that a part or all of the Sb sites (2d) are occupied by Va, respectively.

As reported, α-Mg_3Sb_2 dissolves about 9 mol.% Mg_2Sn in the Mg_2Sn–Mg_3Sb_2 vertical section in Ref. [78]. As shown in Fig. 1 of the Mg–Sn system and Fig. 5 of the Mg–Sb system, it is difficult to form rich Sn or Sb terminal solid solutions on account of the immiscibility with Mg. Thus, the α-Mg_3Sb_2 phase of the Mg–Sn–Sb ternary system is regarded as $Mg_3(Sb, Sn, Va)_2$, which some Sb atoms are substituted by Sn atoms in the second sublattice.

3.4.3. β-Mg_3Sb_2 phase

In the Mg–Sb binary system, the β-Mg_3Sb_2 phase is with the Mn_2O_3 prototype ($Ia\bar{3}$) [85]. Fig. 3 shows the crystal structure of the β-Mg_3Sb_2 phase. Three kinds of atoms are located in the unit cell: Sb1 (8b) at (0.25, 0.25, 0.25), Sb2 (24d) at (0.465, 0, 0.25), Mg (48e) at (0.385, 0.145, 0.38). Compare to Sb1 (8b) atoms, the Sb2 (24d) atoms have higher dense space and shorter distances between the center Sb atom and its coordinate atoms in the asymmetrical unit cell, as shown in Fig. 3(b) and (c). It is reasonable to assume that vacancies prefer to occupying the Sb2 (24d) sites rather than the Sb1 (8b) sites owing to the unsteady force acting on the Sb2 (24d) sites at elevated temperature. Therefore, the β-Mg_3Sb_2 phase is modeled as $Mg_6(Sb,Va)_3Sb_1$ model in Mg–Sb binary system. Besides, the solubility of Mg_2Sn phase in β-Mg_3Sb_2 was experimentally investigated in Ref. [78], and reviewed in Ref. [77]. As the similar way treated in α-Mg_3Sb_2 phase, Mg is hardly admixed with Sb or Sn, thus the β-Mg_3Sb_2 phase in the Mg–Sn–Sb ternary system was modeled as $Mg_6(Sb, Sn, Va)_3(Sb, Sn)_1$, where some Sb atoms substituted by Sn atoms in the second and the third sublattices at random.

3.4.4. Sn_4Sb_3 phase

Shmetterer et al. [53,58] reported the Sn_4Sb_3 phase with a rhombohedral structure belongs to R$\bar{3}$m space group. Later, Gierlotka [59] confirmed the Sn_4Sb_3 phase as a stable phase in Sn–Sb system by first-principles calculations, and reported the formation enthalpy of Sn_4Sb_3 (−29600 J/mol). Although a small solid solution range of Sn_4Sb_3 phase was proposed in Ref. [60], they did not provide any convincing experimental results on the solubility range of the Sn_4Sb_3 phase. In this work, the Sn_4Sb_3 phase is treated as a stoichiometric phase for simplicity.

3.4.5. SnSb phase

The crystal structure of SnSb phase is rhombohedral based on a distortion of the basic NaCl structure [53,58], which can be identified as

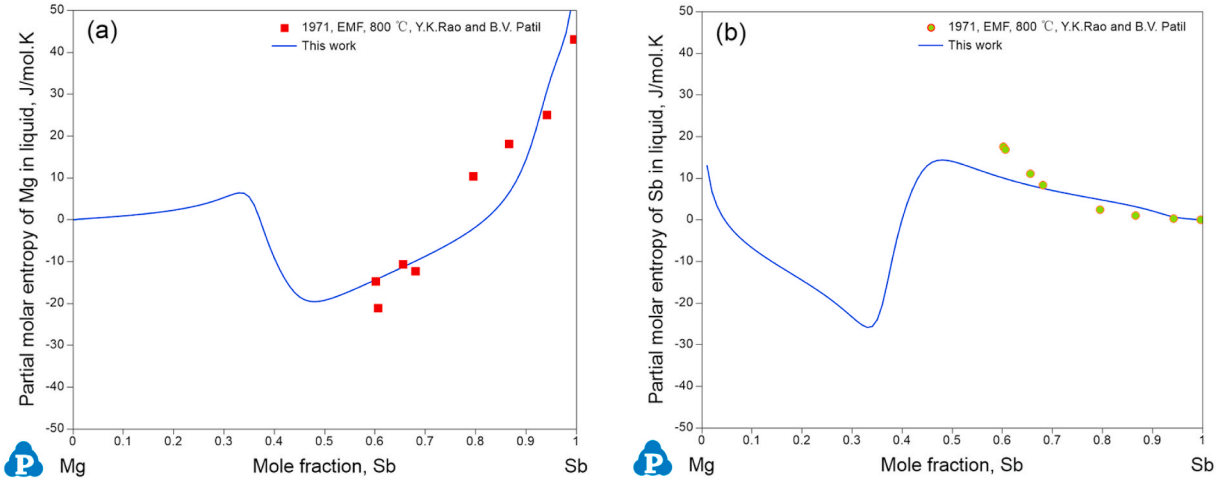


Fig. 9. Calculated partial molar entropy of Mg (a) and Sb (b) in the liquid phase of the Mg–Sb system compared with measured values [45], (Reference state: liquid Mg and liquid Sb).

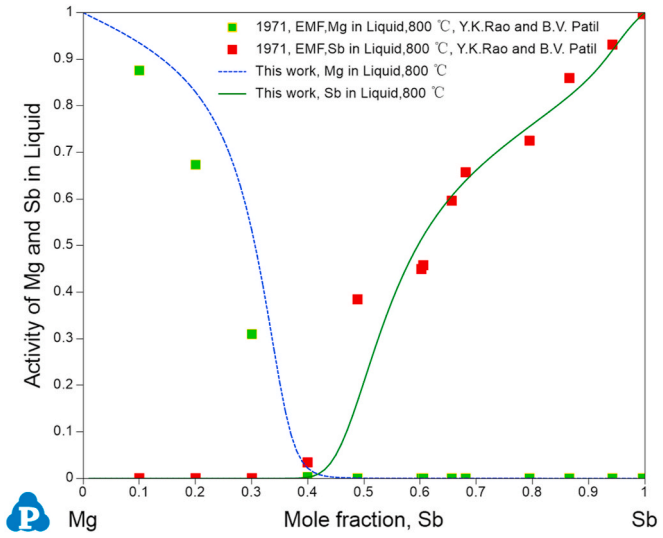


Fig. 10. Calculated activities of Mg and Sb in liquid Mg–Sb solution at 800 °C with the experimental data [45], (Reference state: liquid Mg and liquid Sb).

a mix of Sn_4Sb_3 7-layer building blocks and Sb-layers. In addition, the two-phase area [SnSb + (Sb)] can be interpreted as the lack of relevance among Sn_4Sb_3 building blocks when Sb-layers beyond a certain number. Theoretically, this situation might possibly exist until the complete omission of the Sn_4Sb_3 blocks, which results in pure Sb. On the other hand, once Sb-lays was missed, this situation would give rise to the formation of SnSb with a large homogeneity range and limit of Sn_4Sb_3 by complete omission of the Sb-layers. Therefore, the SnSb phase is modeled as two sublattices $(Sn, Sb, Va)_4Sb_3$ in Sn–Sb binary system, and no Mg solubility in SnSb phase is considered in the Mg–Sn–Sb ternary system as no experimental data is available.

4. Results and discussion

Employing the compatible lattice stabilities of the elements Mg, Sn and Sb compiled by SGTE, the model parameters for various phases in the Mg–Sn–Sb system were obtained in the present work, as shown in Table 2.

4.1. The Mg–Sn binary system

The thermodynamic parameters of the Mg–Sn system were refined based on the description provided by Meng et al. [30]. In order to

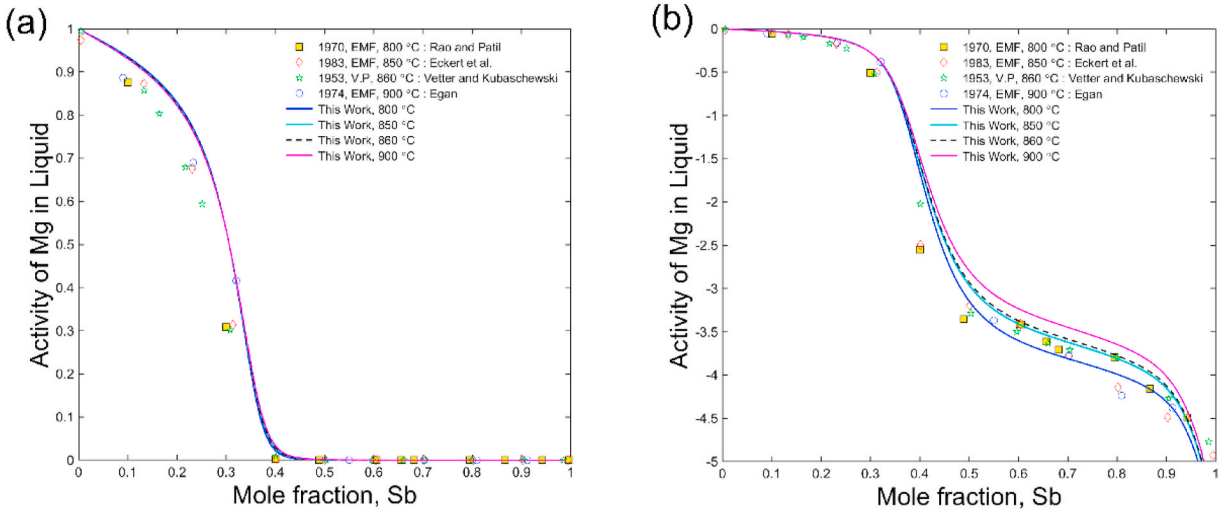


Fig. 11. Calculated activities of Mg in liquid Mg–Sb solution referred to its liquid standard state. (a) Linear scale and (b) logarithm scale.

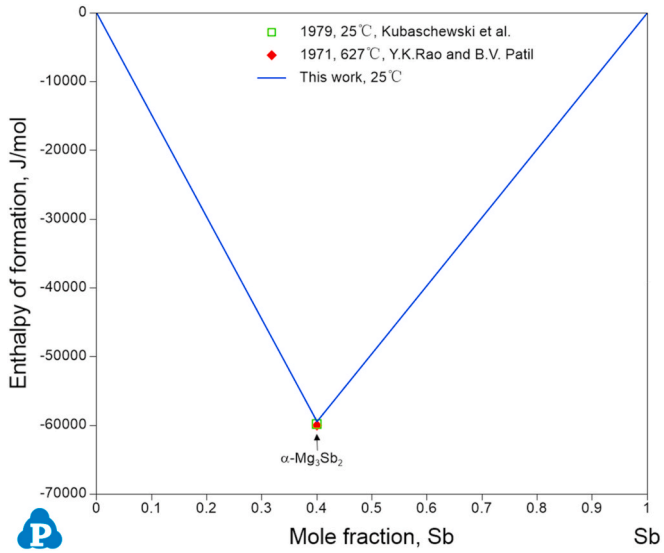


Fig. 12. Calculated enthalpies of formation in comparison with the experimental data [45,47], (Reference state: Hcp-Mg and Rhombohedral-Sb).

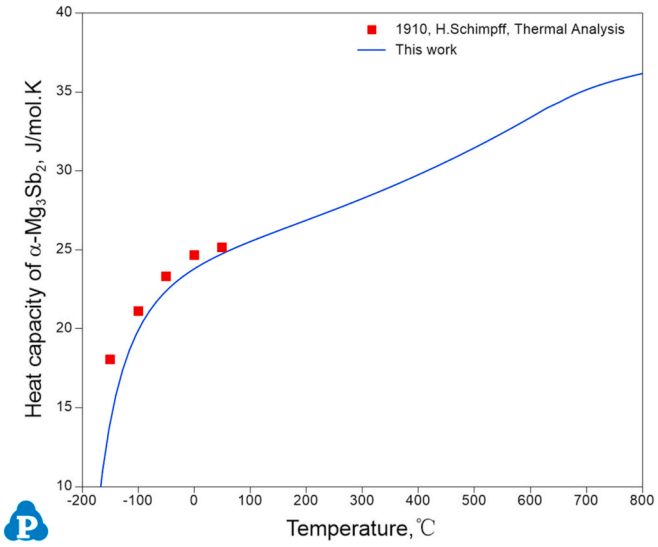


Fig. 13. Calculated heat capacity of α-Mg_3Sb_2 in comparison with the experimental data [52], (Reference state: Hcp-Mg and Rhombohedral-Sb).

Table 5
The comparison of calculated heat and entropy of formation of α-Mg_3Sb_2 phase from solid Mg and Sb with reported values (units: KJ/mol, J/(mol•K)).

T (°C)	Heat of formation [a]	Entropy of formation [a]	Method	Reference
650	−284.7	…	Calorimetry	[48]
	−291.89	−37.45	Calculated	[38]
	−293.78	−49.7	Calculated	This work
627	−299.02 ± 3.14	−59.45 ± 5.02	EMF	[45]
	−291.89	−37.45	Calculated	[38]
	−293.56	−48.97	Calculated	This work
500	−291.89	−37.45	Calculated	[38]
	−292.98	−48.14	Calculated	This work
127 to 277	−238.64 ± 8.37	−15.07 ± 10.47	Calorimetry	[46]
	−292.05 ± 0.6	−47.15 ± 0.8	Calculated	This work
25	−300.19 ± 16.7	−52.50 ± 15.32	Calorimetry	[47]
	−291.89	−37.45	Calculated	[38]
	−291.32	−45.91	Calculated	This work

[a] Entity for the mol is the Mg_3Sb_2 formula unit, and measured with respect to the pure elements at the measured temperature.

properly model the solubility of Sb in Mg_2Sn phase, neutral vacancy was introduced into the model of the Mg_2Sn phase as $(Mg, Va)_2(Sb, Sn)$ for Mg–Sn–Sb ternary system. Hence, a small solubility of $(Mg, Va)_2Sn$ phase in Mg–Sn binary system is also introduced, which is different from the stoichiometric phase in Meng et al. [30]. Eldridge et al. [86] determined the solubility of Mg and Sn in Mg_2Sn with the range of a few tenths of a percent at elevated temperatures. The largest solid solubility of Mg_2Sn in this work is 0.9 at.% Sn, i.e. nine tenths of a percent.

The calculated invariant reactions compared with available experimental results and previous calculations [30,50,87–92] are collected in Table 3. The calculated congruent melting point of Mg_2Sn phase is 771 °C, in good agreement with the experimental results of 770 °C measured by Steiner et al. [88] and Egan [50], which was reviewed as the most reliable result by Nayeb-Hashemi and Clark [31]. The calculated temperature of invariant reaction L ↔ Hcp(Mg) + Mg_2Sn is 561.1 °C, which agrees well with the experimental results 561.2 °C from Ref. [88] and recommended in Ref. [31]. Fig. 4 shows the calculated phase diagram in comparison with experimental data compiled by Nayeb-Hashemi and Clark [31], showing a good agreement.

4.2. The Mg–Sb binary system

Table 4 compares the current calculated invariant reactions with available experimental results and previous calculations [34,38,40–45]. Fig. 5(a) shows the calculated phase diagram of the Mg–Sb binary system, and Fig. 5(b), shows the calculated phase diagram in comparison with the experimental data [42,43]. As can be seen in the Fig. 5 and Table 4, the liquidus temperature and invariant reactions between the calculation and experimental results are in reasonable agreement.

The calculated Gibbs energies and enthalpies of mixing of liquid Mg–Sb alloys are compared with the experimental information [45,49] at different temperatures in Fig. 6 and Fig. 7, respectively. The present calculated results show good agreement with the experimental data except a little deviation of the enthalpy of mixing data reported by Vetter and Kubaschewski using vapor pressure measurement [49]. The reasonable agreement with experimental data also supports the choosing of associate solution model. The Mg_3Sb_2 associate in the liquid phase produced the temperature dependence of enthalpy of mixing as shown in Fig. 7.

The calculated entropy of mixing of liquid Mg–Sb alloys is acceptable compared with experimental values [45,49] within the error range in Fig. 8. The typical M shape lines represent the short-range-ordering tendency at the Mg_3Sb_2 composition, which are also consistent with "V" shape of the enthalpy of mixing curves as shown in Fig. 7. The calculated results show good agreement with the data derived by EMF experimental data [45] but show some deviation with the data reported in Ref. [49], which are derived from vapor pressure results. But we should notice that all these entropy data reported in Refs. [45,49] are actually derived experimental data. The real measured experimental data in Refs. [45,49] are EMF and vapor pressure data, which are equivalent to the activity data. The good agreement with the real experimental data will be shown later.

The comparison of partial molar entropy of Mg and Sb in the liquid phase between the experimental data [45] and the calculated data are shown in Fig. 9. It is clearly seen that the sudden change of the partial molar entropy of Mg and Sb around the composition of Mg_3Sb_2 phase is related to the strong-short-range ordering tendency in liquid phase. As can be seen in the Table 2 and the formula 4 ($^0G^L_{Mg_3Sb_2}$, in expression), the value of a_2 parameter is very negative and gives a major impact on the strong chemical ordering behavior of the liquid phase.

Fig. 10 presents the activities of Mg and Sb in the liquid Mg–Sb alloys with the reported experimental data [45]. The calculated activities of Sb

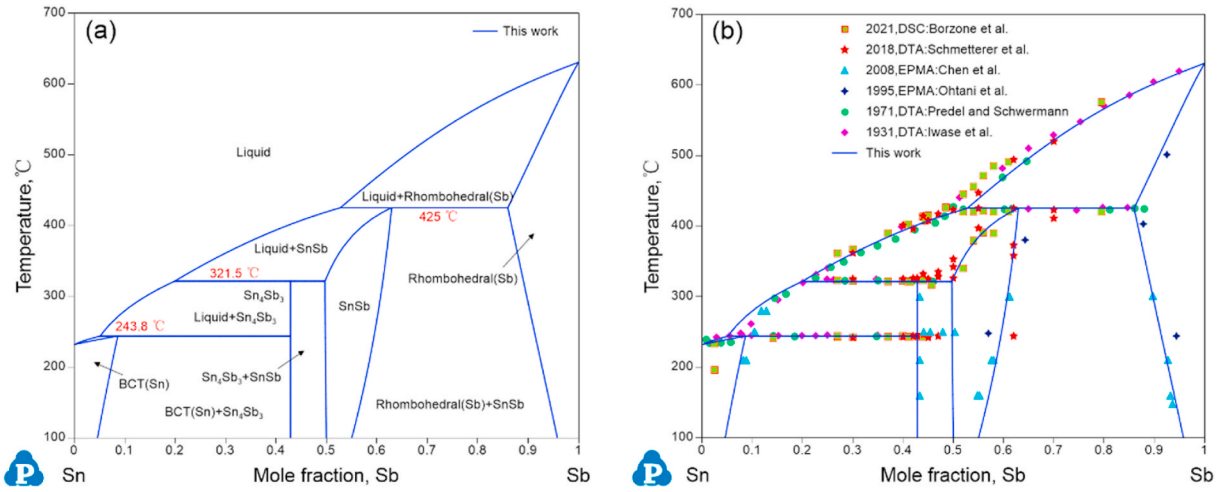


Fig. 14. Calculated phase diagram of the Sn–Sb system in the present work, compared with the experimental data [55,57,58,60,61,67].

Table 6
Invariant reactions of the Sn–Sb system.

Reaction	T (°C)	Composition, x_{Sb}			Method	Reference
L + Rhom(Sb)	425	…	…	…	Experiment	[58]
↔ SnSb	420	…	…	…	Experiment	[60]
	422	…	…	…	Experiment	[68]
	425	…	…	…	Experiment	[67]
	425	0.50	…	0.652	Experiment	[57]
	425	…	…	…	Experiment	[70]
	430	…	…	…	Experiment	[69]
	424.5	0.489	0.869	0.674	Calculated	[59]
	425.0	0.527	0.860	0.629	Calculated	This work
L + SnSb ↔	326	…	…	0.428	Experiment	[58]
Sn_4Sb_3	321	…	0.49	0.43	Experiment	[60]
	325	…	…	…	Experiment	[67]
	324	0.21	…	0.43	Experiment	[57]
	325	…	…	…	Experiment	[70]
	322.6	0.194	0.462	0.428	Calculated	[59]
	321.4	0.199	0.496	0.428	Calculated	This work
L + Sn_4Sb_3 ↔	244	…	0.428	…	Experiment	[58]
Bct(Sn)	243	…	…	…	Experiment	[60]
	244	…	…	…	Experiment	[68]
	246	…	…	…	Experiment	[67]
	250	0.06	…	0.1	Experiment	[57]
	245	…	…	…	Experiment	[70]
	244.2	0.071	0.428	0.098	Calculated	[59]
	243.82	0.052	0.428	0.088	Calculated	This work

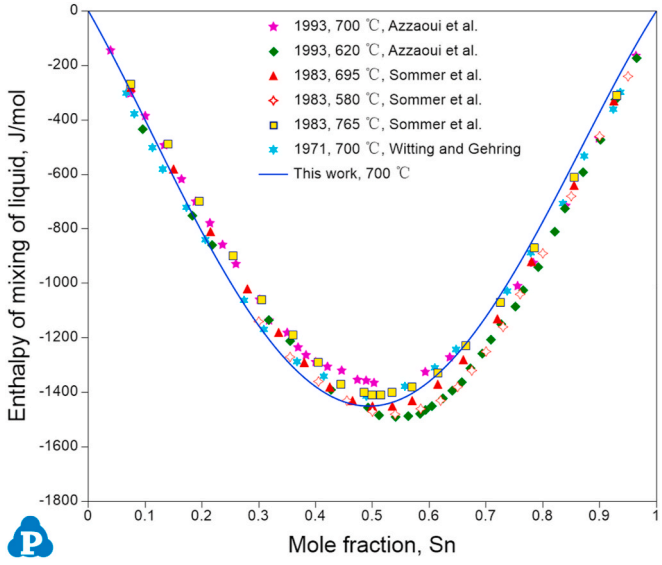


Fig. 15. Calculated enthalpy of mixing of liquid phase together with the experimental data [71–73] at 700 °C, (Reference state: liquid Sn and liquid Sb.).

are in good agreement with the experimental data [34,45], but the activities of Mg show some slight discrepancy. In the original paper [45], we found that they only measured the EMF in the x(Sb) range from 0.4890 to 0.9965, i.e. the activity data of Sb data are the only "real" experimental data, which agree well with the calculated results in this work. The activity data of Mg reported in Refs. [34,45] were actually derived based on Gibbs-Duhem relation.

Fig. 11 compares the calculated activities of Mg in Mg–Sb liquid phase at different temperature with the experimental data [45,49–51]. The calculated results show good agreement with the experimental data from the vapor pressure measurement [49], although the entropy of mixing calculated in this work shows deviation from their derived entropy of mixing reported in Ref. [49] as shown in Fig. 8. That is an example showing that some derived "experimental" data might provide some misleading information, and need to be used carefully.

The enthalpies of formation of α-Mg_3Sb_2 phase in the Mg–Sb binary system at 25 °C are calculated as shown in Fig. 12. The calculated enthalpy of α-Mg_3Sb_2 at 25 °C in this work is −291.32 kJ/mol, in good agreement with the experimental value of −300.19 ± 16.7 kJ/mol reported in Ref. [47]. Table 5 lists the more detailed comparison of calculated enthalpy and entropy of formation of α-Mg_3Sb_2 phase with the previous reported data [38,45–48].

The calculated heat capacity (Cp) of α-Mg_3Sb_2 compared with the experimental data [52] is shown in Fig. 13. In the present work, the T•lnT and T^2 terms are introduced to fit the Cp data [52]. The calculated values agree well with the experimental data except the point at the lowest temperature of −150 °C. As the SGTE data for the Cp of Mg and Sb are only validated at T > 298 K, and the extrapolation to low temperature application needs to be very careful. Therefore, the deviation at low temperature is considered as acceptable, and the assessed thermodynamic parameters of α-Mg_3Sb_2 are reasonable.

4.3. The Sn–Sb binary system

The comparison of calculated phase diagram in this work with the experimental data [55,57,58,60,61,67] is given in Fig. 14. The calculated invariant reactions and comparison between the experimental data and the assessed data [57–60,67–70] are shown in Table 6. The calculated phase diagram shows good agreement with most of the

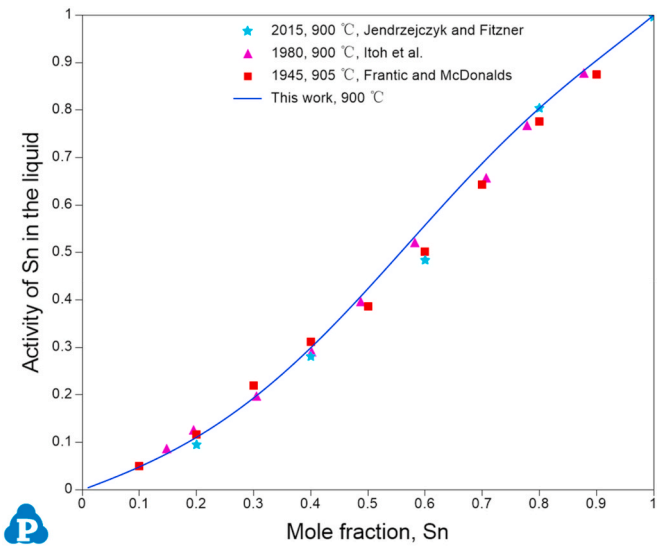


Fig. 16. Calculated activities of Sn in the liquid phase together with the experimental data [74–76] at 900 °C, (Reference state: liquid Sn and liquid Sb).

experimental data on the temperature of the liquidus and the invariant reactions, as well as the solid solubility range of BCT(Sn), Rhombohedral (Sb) and SnSb phase. The significant difference between the current calculated phase diagram with previous assessed phase diagrams is about the Sn_4Sb_3 phase and the Sn_3Sb_2 phase. In the present work, we accepted the dedicated experimental results [58,60] and ab initio calculations [59] in the most recently published papers and treated the Sn_4Sb_3 phase as the only stable intermediate phase in equilibrium with SnSb phase. Although linear equation is used for the interaction parameters of the liquid phase, no inverted miscibility gap appears at high temperatures.

Fig. 15 shows calculated enthalpy of mixing of the liquid phase compared with experimental data given by Witting et al. [71], Azzoui et al. [72], and Sommer et al. [73]. The experimental data [72] show a small difference between 620 °C and 700 °C, but with considering the possible experimental error and the experimental data from other sources [71,73], it is reasonable to treat the enthalpy of mixing as temperature independence by using linear equation for the interaction parameters of the substitutional solution model. The calculated activity of Sn in the liquid of Sn–Sb binary system also agrees well with the experimental data [74–76], as shown in Fig. 16.

4.4. The Mg–Sn–Sb ternary system

The calculated liquid projection of the Mg–Sn–Sb ternary system is provided in Fig. 17. The reaction scheme of the invariant reactions in this ternary system is exhibited in Fig. 18. And the corresponding compositions and temperatures of the invariant reactions in Mg–Sn–Sb ternary system are also given in Table 7. As shown in Fig. 17, eight primary phase regions are observed in this liquidus projection, where Mg_2Sn and α-Mg_3Sb_2 phases composed most of the primary phase regions due to their high stability. The invariant reactions (U2, U3, U4 and E5) can also be observed in the vertical section of Mg0.4Sb0.6-Mg0.4Sn0.6, which will be discussed later, respectively. Moreover, the shape of the liquidus projection calculated in this work is in general consistent with the assessed phase diagram proposed by Jonsson and Agren [81], except in the Sn–Sb binary side, because the stable

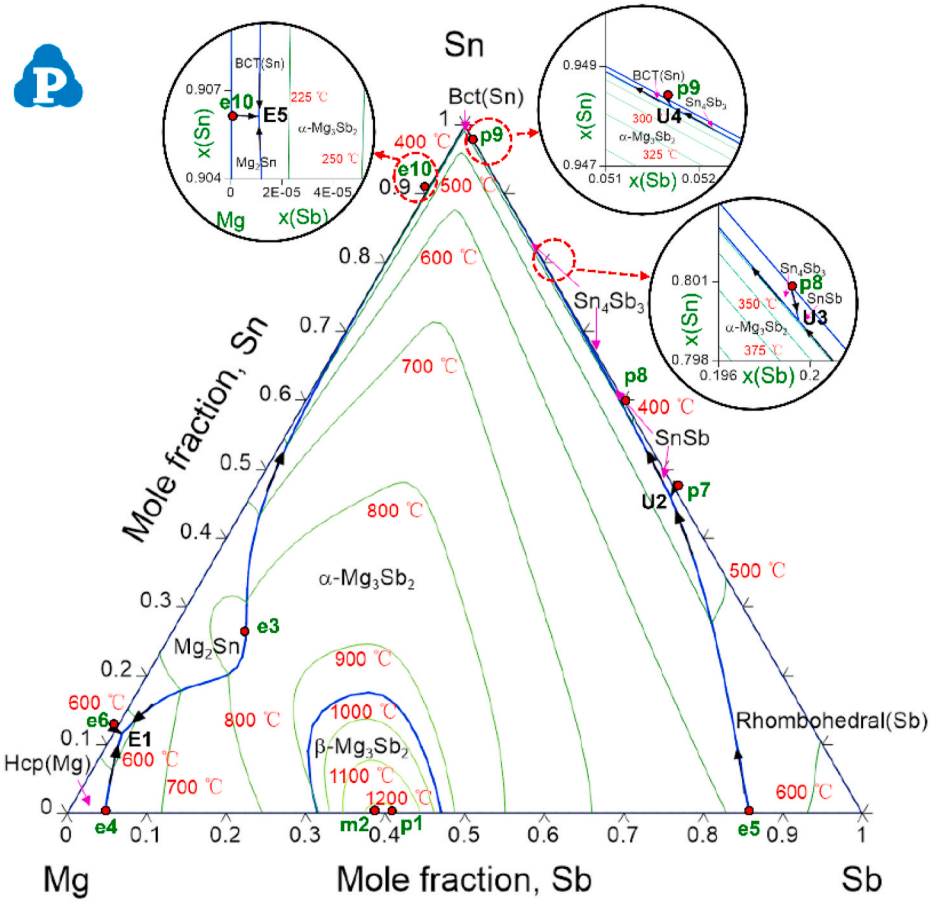


Fig. 17. Calculated liquid projection of the Mg–Sn–Sb ternary system in the present work.

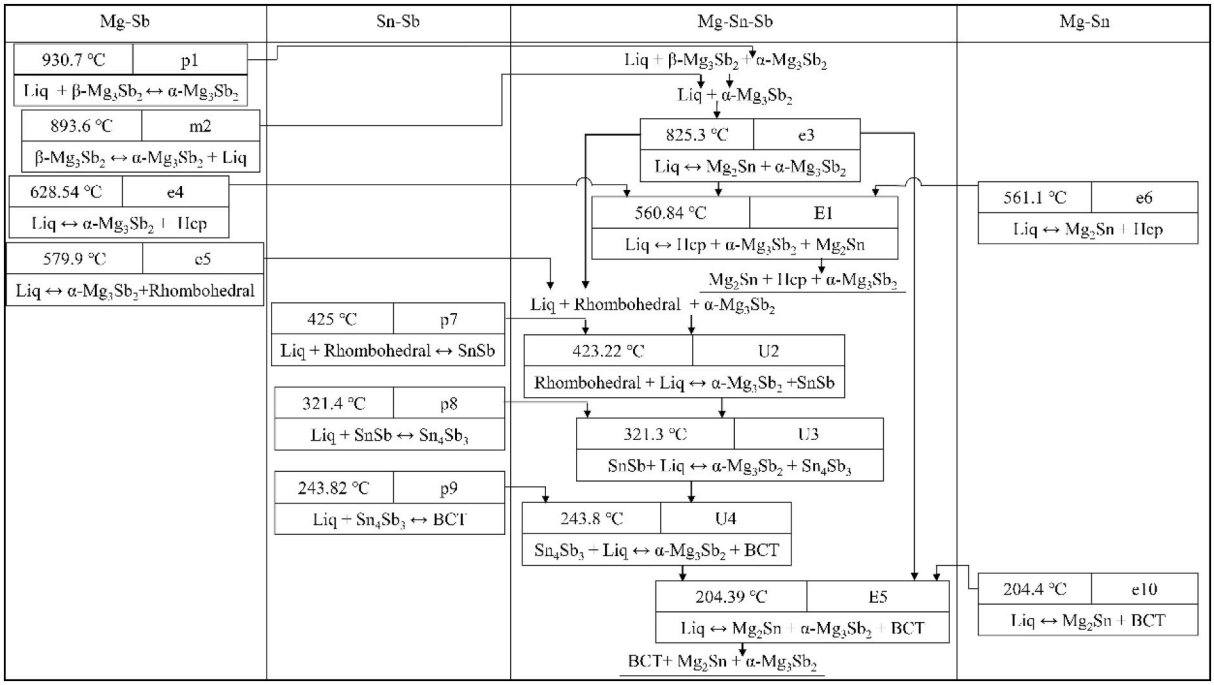


Fig. 18. The reaction scheme of invariant reactions in the Mg–Sn–Sb ternary system.

Table 7
Calculated invariant reactions in the Mg–Sn–Sb ternary system.

Invariant reaction	Type	T (°C)	Composition		Reference
			x_{Mg}^{Liq}	x_{Sb}^{Liq}	
L ↔ Mg_2Sn + α-Mg_3Sb_2	e3	825.3	0.652	0.099	This work
L ↔ Hcp(Mg) + Mg_2Sn + α-Mg_3Sb_2	E1	560.84	0.8736	0.0113	This work
L + Rhom(Sb) ↔ α-Mg_3Sb_2 + SnSb	U2	423.22	0.0117	0.5286	This work
L + SnSb ↔ α-Mg_3Sb_2 + Sn_4Sb_3	U3	321.3	0.00096	0.1995	This work
L + Sn_4Sb_3 ↔ α-Mg_3Sb_2 + Bct(Sn)	U4	243.8	0.00011	0.0517	This work
L ↔ Mg_2Sn + α-Mg_3Sb_2 + Bct(Sn)	E5	204.39	0.094	0.00001	This work

compound Sn_4Sb_3 in the Sn–Sb binary system was not reported at that time.

The calculated activity of Mg in the liquid of Mg–Sn–Sb ternary system at 800 °C is exhibited in Fig. 19, compared with the experimental results at different mole ratio (Sn/Sb) provided by Eckert et al. [51] employing an EMF technique. As can be observed, the calculated results in the present work at 800 °C also are found to be in reasonable agreement with experimental data [51] in the liquid of Mg–Sn–Sb ternary system. Therefore, the tendency of chemical ordering behavior of liquid phase in the Mg–Sn–Sb ternary system, which was reported by Jonsson and Agren [81], could also be described successfully by using the associate solution model. Besides, the calculated results of activity of Mg, Sn and Sb in the liquid of this ternary system with different mole ratio (Sn/Sb) at 800 °C is also provided in Fig. 20.

Fig. 21 shows the calculated isothermal section of the Mg–Sn–Sb system at 300 °C in comparison with the available experimental data reported by Kinzhibalo et al. [80]. The calculated phase diagram in this work shows some deviation from the suggested phase diagram (the red dotted line) in Ref. [80], but is in good agreement with their experiments data points [80]. The Liquid + α-Mg_3Sb_2 + SnSb three phase region reported in Ref. [80] is calculated as the Liquid + α-Mg_3Sb_2 + Sn_4Sb_3 three phase region. The most recently dedicated experimental results in

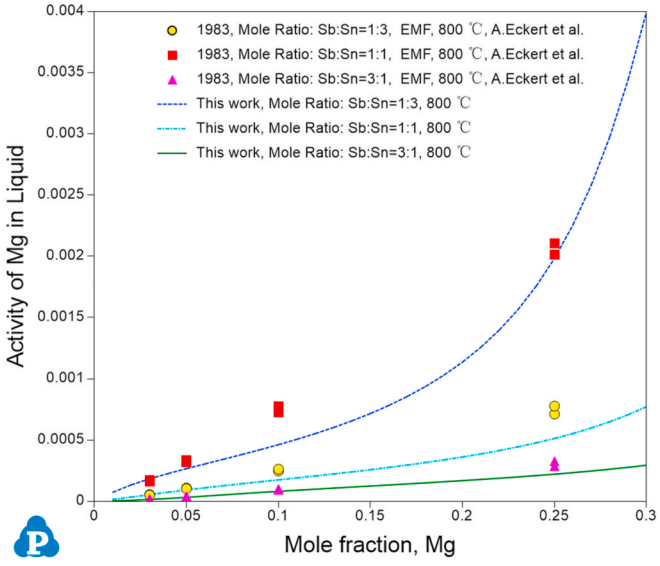


Fig. 19. Comparison between the calculated activity of Mg in the liquid phase and the experimental data [51] with different mole ratio (Sn/Sb) at 800 °C, (Reference state: liquid Mg, liquid Sn and liquid Sb).

Sn–Sb binary sub-system supported the stable of the Sn_4Sb_3 phase at 300 °C [58,60]. The calculation solubility of the third component in Mg_2Sn phase and α-Mg_3Sb_2 phase are less than 5 at.%, in consistent with the experimental results [80].

The calculated vertical section between the compounds Mg_2Sn and Mg_3Sb_2 is compared with the experimental data [78] in Fig. 22. As can be seen, the data reported by Bolshakov et al. [78] are quite well reproduced, except for some points at Mg_3Sb_2 rich side, which are located inside the liquid and β-Mg_3Sb_2 two phase region. The same discrepancies also exist in the thermodynamic modeling provided and discussed in detail by Jonsson and Agren [81], because this vertical section is not a true pseudo-binary system. Besides, the completely solid solubility between the Mg_2Sn and the high temperature β-Mg_3Sb_2 phase

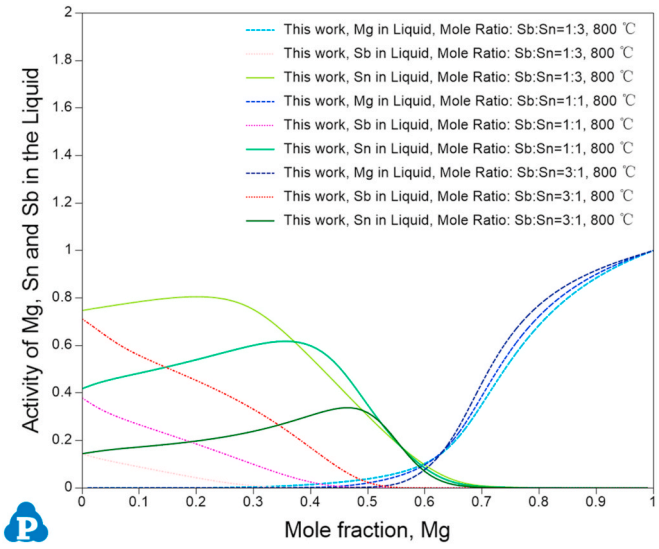


Fig. 20. Calculated activity of Mg, Sn and Sb in the liquid phase of the Mg–Sn–Sb ternary system with different mole ratio (Sn/Sb) at 800 °C in the present work, (Reference state: liquid Mg, liquid Sn and liquid Sb).

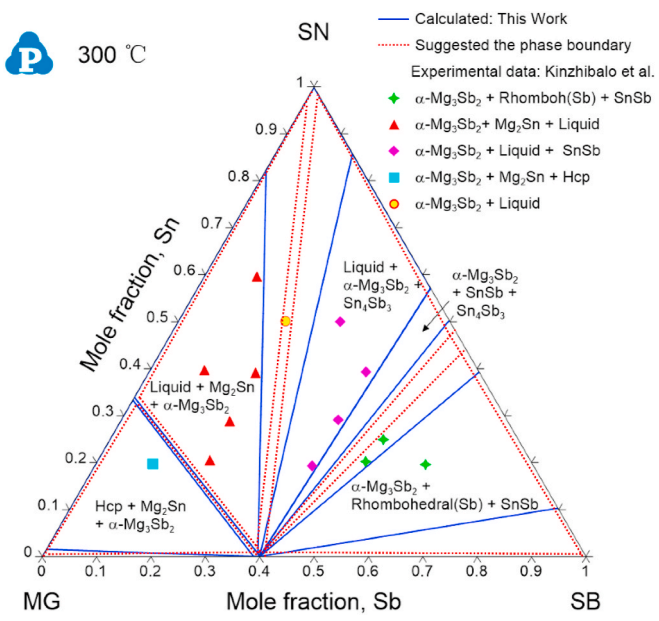


Fig. 21. Isothermal section of the Mg–Sn–Sb ternary system at 300 °C, calculated result in this work compared with the experimental data from Ref. [80] and their suggested phase boundaries.

proposed in Ref. [78] is difficult because of their entirely different crystal structures [77,81] and stable high temperature range, as shown in Figs. 4 and 5(a). The maximum solubility limit of Mg_2Sn phase calculated in the present work is around 25.7 at.% Sn at 780 °C (Fig. 22), while this value reported in Ref. [81] is less than 15 at.% Sn [81] As there is no more reliable experimental data is available, the current agreement with the experimental data [78] in the present work is considered as acceptable with the advantage of all the updated binary sub-systems over the previous thermodynamic modelling [81].

Jonsson and Agren [81] had calculated the vertical section of Mg0.4Sb0.6-Mg0.4Sn0.6 and predicted the isotropic temperature base on the theory of homogeneous nucleation. The vertical section of Mg0.4Sb0.6-Mg0.4Sn0.6 is also calculated in this work and shown in Fig. 23. In this section, the calculated results in the present work show

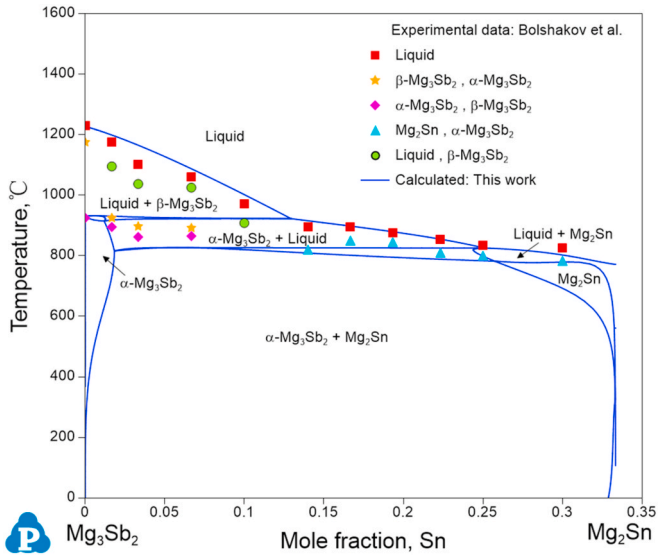


Fig. 22. Calculated vertical section of Mg_3Sb_2–Mg_2Sn with experimental data [78].

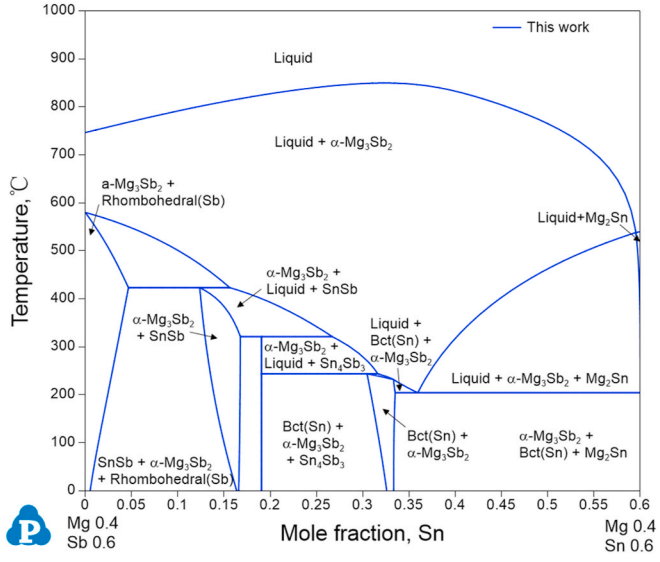


Fig. 23. Calculated vertical section of Mg0.4Sb0.6-Mg0.4Sn0.6 in the present work.

that there are four invariant reactions, U2, U3, U4 and E5, which need to be confirmed in the future experiments. Moreover, these invariant reactions except U3 are very similar to the predicted results in Ref. [81]. As reported from Refs. [58,60], the Sn_4Sb_3 phase is the stable phase. And the Sn_4Sb_3 phase is included rather than the Sn_3Sb_2 phase in this section by directly extrapolating the constituent sub-binary system. In addition, the calculated SnSb phase boundary is not sensitive to the most ternary parameters because this section is very close to phase diagram of the Sn–Sb binary system.

4.5. *Simulation of solidification process*

The effect of Sb additions on the creep behavior of as-cast Mg–5Sn alloys with 0.15, 0.4 and 0.7 wt.% Sb was investigated by Nayyeri and Mahmudi [24]. As they believed [24], the coarse dendritic structure of the base alloy is refined obviously after the addition of Sb, especially this refining effect being more pronounced in tested alloy with 0.4 wt.% Sb, which has the highest creep-resistance. Because the thermally stable

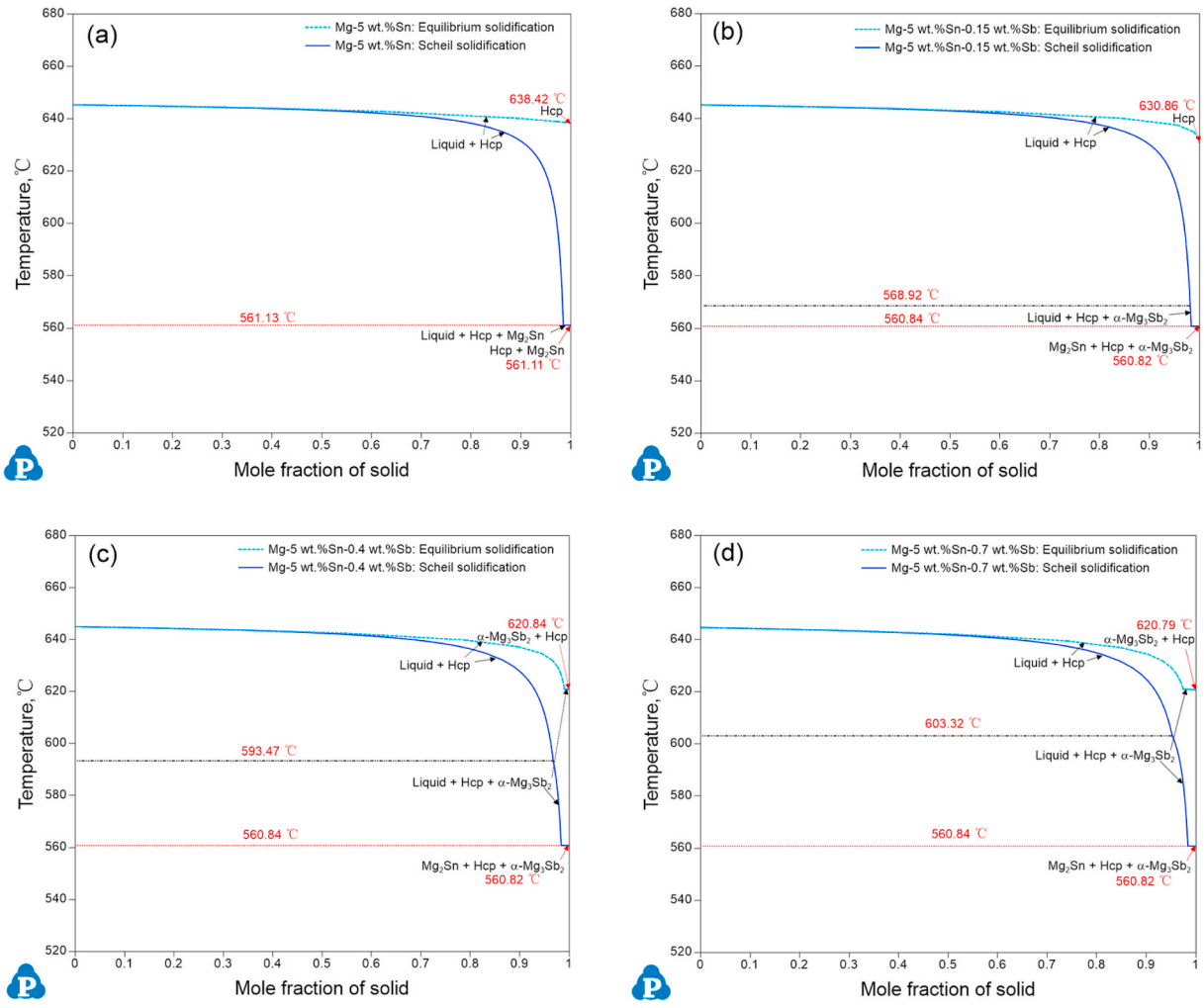


Fig. 24. Calculated solidification path for the four typical alloys.(a) Mg–5Sn alloy, (b) Mg–5Sn-0.15Sb alloy, (c) Mg–5Sn-0.4Sb alloy, (d) Mg–5Sn-0.7Sb alloy.

α-Mg_3Sb_2 and Mg_2Sn particles, which can strengthen both matrix and grain boundaries during high temperature creep deformation, were simultaneously produced in these investigated systems. In other words, the very high thermal stability of these alloy microstructure is believed to be enhanced by the presence of the highly stable α-Mg_3Sb_2 phase. In fact, the different microstructures of the as-cast alloys would be affected by solidification sequences, which may be responsible for the effects of the adding alloying elements. And the real process of permanent mold casting fall in between the Scheil solidification and equilibrium solidification, but is generally much close to the former, which is assumed that there is no diffusion in the solid phase whereas liquid is with fully mixing and fast diffusion. The solidification paths from 750 °C of the four typically alloys reported by Nayyeri and Mahmudi [24] are also simulated by the Scheil solidification and equilibrium solidification, as shown in Fig. 24. The primary phases in these alloys are confirmed to be Hcp(Mg) by SEM micrographs [24] and calculated results in this work. When the temperature of residual liquid is achieved at 560.84 °C, the ternary eutectic reaction, Liquid ↔ Hcp + α-Mg_3Sb_2 + Mg_2Sn, comes up until the liquid phase disappear. The calculated results agree reasonably with the SEM observation of eutectic structure (lamellar of α-Mg_3Sb_2 and isolated spherical Mg_2Sn particles) in Ref. [24], which can intensify grain boundary and retard dynamic recrystallization. Besides, the α-Mg_3Sb_2 phase can directly form from the liquid phase during the solidification process beyond the critical content of 0.2 wt.% Sb. And the volume fraction of the more thermally stable α-Mg_3Sb_2 increases from liquid phase at more elevated temperature with the many amounts of Sb, but the latent heat of crystallization of α-Mg_3Sb_2 phase might partially counteract the effect of the undercooling during solidification. Hence, as reported by Nayyeri and Mahmudi [24], the microstructure of Mg–5Sn-0.4Sb alloy may be the optimal, which can positively improve the high-temperature creep-resistance behavior.

5. Conclusions

The thermodynamic description of Mg–Sn–Sb ternary system has been developed based on the re-optimization of Mg–Sb and Sn–Sb binary systems together with a revised Mg–Sn binary system. For the Mg–Sb binary system, the associate solution model was applied to describe the short-range ordering behavior of the liquid phase and avoided the inverted miscibility gap at high temperature. For the Sn–Sb binary system, the most recently published experimental data on the stability of the Sn_4Sb_3 phase are accepted in the optimization. The Mg–Sn binary system was slighted revised from previous publication in order to produce reasonable Mg–Sn–Sb ternary phase diagram. The liquidus projection, isothermal section and several vertical sections of this ternary system have been calculated. The results showed reasonable agreement with all available experimental data and present a similar or better agreement with experimental data than the only previous thermodynamic modeling. Thermodynamic study and re-assessment of the Mg–Sn–Sb ternary system can be integrated into the Mg alloys thermodynamic database for developing alloy-type anode materials in rechargeable magnesium batteries.

Data availability

The raw/processed data required to reproduce these findings are already available in the supplementary.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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